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Wronskian of the function $y_c = c_1 + c_2 \cos x + c_3 \sin x$ is

Answer (Please select your correct option)

☐ 0

☐ 1

☐ 2

☐ 3

correct answer solve
by hadi



The value of amplitude in the solution $X=40\sin(7t+4)$ is

Answer (Please select your correct option)

☐ 7

☐ 4

☐ 40

☐ 20

correct answer solve
by hadi

Made By: Waqar Siddhu

The degree of the differential equation $[1 - x(\frac{dy}{dx})^2]^{\frac{3}{2}} = \frac{d^2y}{dx^2}$ is

Answer (Please select your correct option)

☐ 1

☐ 2

☐ 3

☐ None of these

correct answer solve
by hadi

Which of the following is the equation of pendulum?

Answer (Please select your correct option)

☐ $\frac{d^2\theta}{dt^2} + \frac{g}{l} \sin \theta = F(t)$

correct answer solve
by hadi

☐ $\frac{d\theta}{dt} + \frac{g}{l} \sin \theta = F(t)$

☐ $\frac{du}{dt} = F(u)G(t)$

☐ None of these



The differential equation $(x^2 - 2x + 2y^2) dx + 2xy dy = 0$ is a/an -----differential equation.

Answer (Please select your correct option)



Exact



Non-exact

correct answer solve
by hadi



Homogenous



Separable

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In a Bernoulli equation $\frac{dy}{dx} + \frac{1}{x}y = xy^2$, identify $p(x)$, $q(x)$ & n respectively.

Answer (Please select your correct option)

☐ $\frac{1}{x}, x$ & 2

correct answer solve
by hadi

☐ y, x^2y & 2

☐ $y, \frac{1}{x}$ & 2

☐ $\frac{1}{y}, y$ & 2

If a differential equation is of the form $y f(x, y) dx + x g(x, y) dy = 0$ with $xM - yN \neq 0$, then the Integrating factor is-----.

Answer (Please select your correct option)

☐ $\mu = \frac{1}{yN - xM}$

☐ $\mu = \frac{1}{xM + yN}$

☐ $\mu = \frac{1}{xM - yN}$

correct answer solve
by hadi

☐ $\mu = \frac{1}{yM + xN}$

If the given differential equation is homogeneous and $xM + yN \neq 0$, then the integrating factor of the given differential equation is given by-----.

Answer (Please select your correct option)

☐ $u = \frac{1}{xM + yN}$

correct answer solve
by hadi

☐ $u = \frac{1}{xM - yN}$

☐ $u = \exp \left[\int \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} dy \right]$

☐ $u = \exp \left[\int \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} dx \right]$

Number of constant solutions obtained from the separable differential equation $\frac{dy}{dx} = \frac{y^3 - 1}{x}$ is---

Answer (Please select your correct option)

☐

0

correct answer solve
by hadi

☐

1

☐

2

☐

3

If $\frac{dy}{dx} = x$, then $y = \dots\dots\dots$

Answer (Please select your correct option)

☐ $y = \frac{x^2}{2} + c$

correct answer solve
by hadi

☐ $y = x^2 + c$

☐ $y = x + c$

☐ $y = \frac{x}{2} + c$

To convert the non-homogenous differential equation $\frac{dy}{dx} = \frac{2x+3y-4}{4x+6y+9}$ into variable separable form, we use the substitution -----.

Answer (Please select your correct option)

☐ $Z = x + y$

☐ $Z = 4x + 3y$

☐ $Z = 2x + 6y$

☐ $Z = 2x + 3y$

correct answer solve
by hadi

Made By: Waqar Siddhu

In exponential model for the population growth $P(t) = P_0 e^{kt}$ If $k < 0$, then $\lim_{t \rightarrow \infty} P(t) = \dots$

Answer (Please select your correct option)

☐ ∞ ☐ 0☐ $-\infty$ ☐ 1

correct answer solve
by hadi... $k > 0$ then
infinte

In exponential model for the population growth $P(t) = P_0 e^{kt}$ If $k > 0$, then $\lim_{t \rightarrow \infty} P(t) = \dots$

Answer (Please select your correct option)

☐ ∞

correct answer solve
by hadi

☐ $-\infty$

☐ 0

☐ 1

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are-----.

Answer (Please select your correct option)

☐ P=0, P= b/a

☐ P=0, P=a/b

correct answer solve
by hadi

☐ P=a/b, P=b/a

☐ P=0, P=0

If $f(x)$ and $g(x)$ are linearly dependent on I , then-----.

Answer (Please select your correct option)

☐

$W(f,g)(x) \neq 0$ (Wronskian) for all x in the interval I .

☐

$W(f,g)(x) = 0$ (Wronskian) for all x in the interval I .

correct answer solve
by hadi

☐

$W(f,g)(x)$ may or may not be zero for all x in the interval I .

☐

$W(f,g)(x)$ is not defined for all x in the interval I .

What is annihilator operator of the function $g(x) = \sin x$?

Answer (Please select your correct option)

☐ $(D^3 + 1)$

☐ D^2

☐ $D^2 - 1$

☐ $D^2 + 1$

correct answer solve
by hadi

Made By: Waqar Siddhu

Which of the following is the annihilator operator of the function $g(x) = 4x - 5$?

Answer (Please select your correct option)

☐

D^2

correct answer solve
by hadi

☐

$(D + 4)$

☐

$(D^3 - 4)$

☐

$(D - 4)$

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

☐ $\frac{dy}{dx} = -\frac{x}{y}$

☐ $\frac{dy}{dx} = \frac{x}{y}$

☐ $\frac{dy}{dx} = -\frac{y}{x}$

☐ $\frac{dy}{dx} = \frac{y}{x}$

correct answer solve
by hadi

Made By: Waqar Siddhu



If $y = 1 + 6x^2 - 7x^3$, then which of the following is true for it ?

Answer (Please select your correct option)



Its annihilator operator is D .



Its annihilator operator is D^2 .



Its annihilator operator is D^3 .



Its annihilator operator is D^4 .

correct answer solve
by hadi

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If $y_1 = x^2$ and $y_2 = x$ are the first and second solution of $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ $W(y_1, y_2)$ must be equal to zero on the indicated interval.

☐ $W(y_1, y_2)$ must be equal to non-zero on the indicated interval.

correct answer solve
by hadi

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correct answer solve
by hadi

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by hadi

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Made By: Waqar Siddhu



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Exact

☐

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correct answer solve
by hadi

☐

Homogenous

☐

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correct answer solve
by hadi

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correct answer solve
by hadi

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If the given differential equation is homogeneous and $xM + yN \neq 0$, then the integrating factor of the given differential equation is given by-----.

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correct answer solve
by hadi

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Number of constant solutions obtained from the separable differential equation $\frac{dy}{dx} = \frac{y^3 - 1}{x}$ is----

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correct answer solve
by hadi

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☐ 2

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Made By: Waqar Siddhu



If $\frac{dy}{dx} = x$, then $y = \dots\dots\dots$

Answer (Please select your correct option)

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correct answer solve
by hadi

☐ $y = x^2 + c$

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To convert the non-homogenous differential equation $\frac{dy}{dx} = \frac{2x+3y-4}{4x+6y+9}$ into variable separable form, we use the substitution -----.

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☐ $Z = 2x + 6y$

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correct answer solve
by hadi

Made By: Waqar Siddhu



In exponential model for the population growth $P(t) = P_0 e^{kt}$ If $k < 0$, then $\lim_{t \rightarrow \infty} P(t) = \dots$

Answer (Please select your correct option)

☐ ∞ ☐ 0

correct answer solve
by hadi

☐ $-\infty$ ☐ 1



In exponential model for the population growth $P(t) = P_0 e^{kt}$ If $k > 0$, then $\lim_{t \rightarrow \infty} P(t) = \dots$

Answer (Please select your correct option)

 ∞

correct answer solve
by hadi

 $-\infty$ 

0



1

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The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are----

Answer (Please select your correct option)



$P=0, P= b/a$



$P=0, P=a/b$

correct answer solve
by hadi



$P=a/b, P=b/a$



$P=0, P=0$



If $f(x)$ and $g(x)$ are linearly dependent on I , then-----.

Answer (Please select your correct option)

☐

$W(f,g)(x) \neq 0$ (Wronskian) for all x in the interval I .

☐

$W(f,g)(x) = 0$ (Wronskian) for all x in the interval I .

correct answer solve
by hadi

☐

$W(f,g)(x)$ may or may not be zero for all x in the interval I .

☐

$W(f,g)(x)$ is not defined for all x in the interval I .

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What is annihilator operator of the function $g(x) = \sin x$?

Answer (Please select your correct option)

☐ $(D^3 + 1)$

☐ D^2

☐ $D^2 - 1$

☐ $D^2 + 1$

correct answer solve
by hadi

Made By: Waqar Siddhu



Which of the following is the annihilator operator of the function $g(x) = 4x - 5$?

Answer (Please select your correct option)

☐ D^2

correct answer solve
by hadi

☐ $(D + 4)$

☐ $(D^3 - 4)$

☐ $(D - 4)$

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The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

☐ $\frac{dy}{dx} = -\frac{x}{y}$

☐ $\frac{dy}{dx} = \frac{x}{y}$

☐ $\frac{dy}{dx} = -\frac{y}{x}$

☐ $\frac{dy}{dx} = \frac{y}{x}$

correct answer solve
by hadi

Made By: Waqar Siddhu



If $y = 1 + 6x^2 - 7x^3$, then which of the following is true for it ?

Answer (Please select your correct option)

☐

Its annihilator operator is D .

☐

Its annihilator operator is D^2 .

☐

Its annihilator operator is D^3 .

☐

Its annihilator operator is D^4 .

correct answer solve
by hadi

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If $y_1 = x^2$ and $y_2 = x$ are the first and second solution of $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ $W(y_1, y_2)$ must be equal to zero on the indicated interval.

☐ $W(y_1, y_2)$ must be equal to non-zero on the indicated interval.

correct answer solve
by hadi

☐ $W(y_1, y_2)$ may or may not be equal to zero on the indicated interval.

☐ $W(y_1, y_2)$ may or may not be equal to non-zero on the indicated interval.

Made By: Waqar Siddhu

Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

The solutions $y_1 = c_1$, $y_2 = c_2 \cos x$, $y_3 = c_3 \sin x$ of a Differential Equation are

Answer (Please select your correct option)

☒ Linearly independent☐ Linearly dependent☐ both of the other options

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1



Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

Wronskian of the functions $y_1(x) = 1$, $y_2(x) = \cos x$, $y_3(x) = \sin x$ is

Answer (Please select your correct option)

☐ 0☒ 1☐ 2☐ 3

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2



Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

The value of amplitude in the solution $X=7\sin(3t+2.45)$ is

Answer (Please select your correct option)

☒ 7☐ 3☐ 2.45☐ 1

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3



Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

The order of the differential equation $\left(\frac{du}{dt}\right)^{\frac{1}{3}} = \frac{1}{u}$ is

Answer (Please select your correct option)

☒ 1☐ 2☐ 3☐ 4

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4



Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

The solution of the differential equation $\frac{dy}{dx} = \tan x$ is

Answer (Please select your correct option)

☐ $y = \ln |\sin x| + c$

☐ $y = \ln |\sec x| + c$

☒ $y = \ln |\operatorname{cosec} x| + c$

☐ $y = \ln |\cos x| + c$

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5



Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

A differential equation $M(x, y) dx + N(x, y) dy = 0$ is said to be an exact if -----.

Answer (Please select your correct option)

☐ $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

☐ $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

☒ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

☐ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

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6



Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

The differential equation $(x^2 - 2x + 2y^2) dx + 2xy dy = 0$ is a/an -----differential equation.

Answer (Please select your correct option)

- ☐ Exact
- ☒ Non-exact
- ☐ Homogenous
- ☐ Separable

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7



Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min) 

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t^2$; the integrating factor will be -----.

Answer (Please select your correct option)

☐ $\mu = -\frac{1}{t}$

☒ $\mu = \frac{1}{t}$

☐ $\mu = t$

☐ $\mu = 2t$

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8



Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n$ for $n \neq 0, 1$ is called a/an ----.

Answer (Please select your correct option)

☐ Non-Exact differential equation☐ Linear differential equation☒ Bernoulli equation☐ Exact differential equation

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9



Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min) 

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^x$$

into linear differential equation, we choose -----

Answer (Please select your correct option)

☐ $v = y^{x-1}$

☒ $v = y^{1-x}$

☐ $v = y^x$

☐ $v = y'$

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10 

Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min) 

To convert the non-homogeneous differential equation $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$ into separable form, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then we use the substitution-----.

Answer (Please select your correct option)

☐ $x = X + h$

☐ $y = Y + h$

☒ $z = a_1x + b_1y$

☐ $z = Y + X$

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11 

Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

In exponential model for the population growth $P(t) = P_0 e^{kt}$ If $k > 0$, then $\lim_{t \rightarrow \infty} P(t) = \dots$

Answer (Please select your correct option)

☒ ∞ ☐ $-\infty$ ☐ 0☐ 1

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12



Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are-----

Answer (Please select your correct option)

☐ P=0, P= b/a☒ P=0, P=a/b☐ P=a/b, P=b/a☐ P=0, P=0

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13



Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min) 

The orthogonal trajectory to the family of curves $3x + 4y = c$ is-----.

Answer (Please select your correct option)

☐ $y = 2x$

☐ $y - x = c$

☒ $4x - 3y = c$

☐ $4x + 3y = c$

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14 

Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

If the tangent lines of two curves are perpendicular at their point of intersection then both the curves are-----

Answer (Please select your correct option)

☐ Non-intersecting curves☐ Parallel curves☒ Orthogonal curves☐ Intersecting curves

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15



Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that $K=0.043$?

Answer (Please select your correct option)

☐ 371.415☐ 380.560☒ 363.279

The formula is $A(t)=A_0 e^{-kt} = 100(e^{-30 \cdot 0.043}) = 363.279$

☐ 360.351

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16



Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

If $y = 1 + 6x^2 - 7x^3$, then which of the following is true for it ?

Answer (Please select your correct option)

☐ Its annihilator operator is D .☐ Its annihilator operator is D^2 .☐ Its annihilator operator is D^3 .☒ Its annihilator operator is D^4 .

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17



Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

If $y = e^{-6x}$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^2$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^3$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)$.

☒ The annihilator operator of $y = e^{-6x}$ is $(D + 6)$.

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18



Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

If $y_1 = x$ and $y_2 = xe^x$ are the first and second solution of $x^2 \frac{d^2 y}{dx^2} - x(x+2) \frac{dy}{dx} + (x+2)y = 0$ on $(0, \infty)$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly dependent.☒ $y_1 = x$ and $y_2 = xe^x$ must be linearly independent.☐ $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly independent.☐ $y_1 = x$ and $y_2 = xe^x$ must be linearly dependent.

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19



Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

If $y_2 = x^3$ is the second solution of $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- ☐ The first solution say y_1 must be non-trivial on the indicated interval.
- ☐ The first solution say y_1 can be trivial on the indicated interval.
- ☐ The first solution say y_1 must be discontinuous on the indicated interval.
- ☐ The first solution say y_1 may or may not be continuous on the indicated interval.

correct answer solve
by hadi

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20





The form of the exact solution to

$$2\frac{dy}{dx} + 3y = e^{-x}, y(0) = 5$$

is

Answer (Please select your correct option)

☐

$$Ae^{-1.5x} + Bxe^{-x}$$

☐

$$Ae^{1.5x} + Be^{-x}$$

☐

$$Ae^{1.5x} + Bxe^{-x}$$

☐

$$Ae^{-1.5x} + Be^{-x}$$

correct answer solve
by hadi

Made By: Waqar Siddhu

The differential equation $2\frac{dy}{dx} + x^2y = 2x + 3, y(0) = 5$ is

Answer (Please select your correct option)

☐ Linear

correct answer solve
by hadi

☐ Nonlinear

☐ Linear with fixed constants

☐ Undeterminable to be linear or nonlinear



The value of amplitude in the solution $X=40\sin(7t+4)$ is

Answer (Please select your correct option)

☐ 7

☐ 4

☐ 40

☐ 20

correct answer solve
by hadi

Made By: Waqar Siddhu

The order of the differential equation $(\frac{\partial^2 u}{\partial t^2})^3 + \frac{\partial u}{\partial t} = 0$ is

Answer (Please select your correct option)

☐ 0

☐ 1

☐ 2

☐ 3

correct answer solve
by hadi

Made By: Waqar Siddhu

Which of the following is the linear differential equation?

Answer (Please select your correct option)

☐ $x \frac{dy}{dx} + y = 0$

correct answer solve
by hadi

☐ $y \frac{dy}{dx} + x = 0$

☐ $x \frac{d^2y}{dx^2} + \frac{dy}{dx} + y^2 = 0$

☐ $\frac{dy}{dx} = e^y$

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A solution obtained by giving particular values to the arbitrary constants in the General Solution of a differential equation is called a -----

Answer (Please select your correct option)

☐ Singular solution

☐ Particular solution

☐ Explicit Solution

☐ None of these

correct answer solve
by hadi

Made By: Waqar Siddhu

Which of the following equations satisfy the differential equation $\frac{dy}{dx} = x$

Answer (Please select your correct option)



$$2y = x^2 + c$$

correct answer solve
by hadi



$$y = x^2 + c$$



$$y = x^2$$



$$y = x + c$$

Made By: Waqar Siddhu



A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n$ for $n \neq 0, 1$ is called a/an ----.

Answer (Please select your correct option)



Non-Exact differential equation



Linear differential equation



Bernoulli equation

correct answer solve
by hadi



Exact differential equation

Made By: Waqar Siddhu

In a Bernoulli equation $\frac{dy}{dx} + \frac{1}{x}y = xy^2$, identify $p(x)$, $q(x)$ & n respectively.

Answer (Please select your correct option)

☐ $\frac{1}{x}, x \text{ \& } 2$

correct answer solve
by hadi

☐ $y, x^2y \text{ \& } 2$ ☐ $y, \frac{1}{x} \text{ \& } 2$ ☐ $\frac{1}{y}, y \text{ \& } 2$

Made By: Waqar Siddhu

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^x$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)



$$v = y^{x-1}$$



$$v = y^{1-x}$$

correct answer solve
by hadi



$$v = y^x$$



$$v = y'$$

Made By: Waqar Siddhu

The differential equation ----- is separable.

Answer (Please select your correct option)

☐ $x(x+y) \frac{dy}{dx} = 4$

☐ $\frac{dy}{dx} = \frac{x^2}{x+xy}$

correct answer solve
by hadi

☐ $\frac{dy}{dx} = \frac{y}{1+xy^3}$

☐ $\frac{dy}{dx} = \frac{xy+3}{1+2xy}$



If $y = x$, then $\frac{dy}{dx} = \dots\dots\dots$

Answer (Please select your correct option)



0



1

correct answer solve
by hadi



x



$x+1$

Made By: Waqar Siddhu

If $f(x)$ and $g(x)$ are linearly dependent on I , then-----.

Answer (Please select your correct option)

- ☐ $W(f,g)(x) \neq 0$ (Wronskian) for all x in the interval I .
- ☐ $W(f,g)(x) = 0$ (Wronskian) for all x in the interval I .
- ☐ $W(f,g)(x)$ may or may not be zero for all x in the interval I .
- ☐ $W(f,g)(x)$ is not defined for all x in the interval I .

correct answer solve
by hadi



Wronskian of $\sin x, \cos x$ is----

Answer (Please select your correct option)

☐ $\sin x$

☐ 1

☐ -1

☐ $\cos x$

correct answer solve
by hadi

Made By: Waqar Siddhu



The orthogonal trajectory to the family of curves $3x + 4y = c$ is-----.

Answer (Please select your correct option)



$$y = 2x$$



$$y - x = c$$



$$4x - 3y = c$$

correct answer solve
by hadi



$$4x + 3y = c$$

Made By: Waqar Siddhu

If T denotes the half life and the initial amount of the substance be A_0 , then the amount of radioactive substance at time T is given by-----.

Answer (Please select your correct option)

☐

$$A(T) = \frac{A_0}{2}$$

correct answer solve
by hadi

☐

$$A(T) = \frac{T}{2}$$

☐

$$A(T) = A_0$$

☐

$$A(T) = \frac{kT}{2}$$

Made By: Waqar Siddhu

If the auxiliary equation has roots $-2, -2$, then the general solution of the differential equation is _____

Answer (Please select your correct option)

☐ $y = c_1 e^{-2x} + c_2 e^{-2x}$

☐ $y = c_1 e^{-2+2x}$

☐ $y = c_1 e^{2x} + c_2 e^{-2x}$

☐ $y = (c_1 + c_2 x) e^{-2x}$

correct answer solve
by hadi

Made By: Waqar Siddhu



If the auxiliary equation has complex roots $-2 \pm 3i$, then the general solution of the differential equation is _____

Answer (Please select your correct option)

☐ $y = ce^{2x} (\cos 10x + \sin 10x)$

☐ $y = e^{-2x} (c_1 \cos 3x + c_2 \sin 3x)$

correct answer solve
by hadi

☐ $y = e^{ix} (c_1 \cos 2x + c_2 \sin 2x)$

☐ $y = e^{2x} (c_1 \cos x - c_2 \sin x)$

If $y = e^{-6x}$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^2$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^3$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D + 6)$.

correct answer solve
by hadi

Made By: Waqar Siddhu

If the complementary solution of $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3\sin x$ is $c_1 e^{-x} + c_2 e^{2x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

☐ The general form of the particular solution will be $(A \sin x + B \cos x)e^x$.

☐ The general form of the particular solution will be $Ax \sin x + Bx \cos x$.

correct answer solve
by hadi

☐ The general form of the particular solution will be $(Ax \sin x + Bx \cos x)e^{-x}$.

☐ The general form of the particular solution will be $A \sin x + B \cos x$.

The form of the exact solution to

$$2 \frac{dy}{dx} + 3y = e^{-x}, y(0) = 5$$

is

Answer (Please select your correct option)

☐

$$Ae^{-1.5x} + Bxe^{-x}$$

☐

$$Ae^{1.5x} + Be^{-x}$$

☐

$$Ae^{1.5x} + Bxe^{-x}$$

☐

$$Ae^{-1.5x} + Be^{-x}$$

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min) The Period of oscillator in the solution $X=50\sin(20t+8.5)$ is

Answer (Please select your correct option)

☐ 0.17643☐ 0.32045☐ 0.31400☐ 0.58000

Start Time: 6:56 AM

58:00

Time Left 2 



The Period of oscillator in the solution $X=50\sin(20t+8.5)$ is

Answer (Please select your correct option)



0.17643



0.32045



0.31400



0.58000



A problem of the differential equation associated with a single value of the unknown function at a given point in the domain of the solution is known as -----

Answer (Please select your correct option)

☐ Boundary Value Problem

☐ Initial Value Problem

☐ None of these

correct answer solve
by hadi

Made By: Waqar Siddhu

The equation $\frac{dy}{dx} = \frac{x(x+1)}{y(y-1)}$ is a/an -----

Answer (Please select your correct option)

☐ partial differential equation.

correct answer solve
by hadi

☐ ordinary differential equation.

☐ polynomial equation.

☐ transcendental equation.

Made By: Waqar Siddhu

Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min) 

If $f(x,y)=0$, then $\frac{dy}{dx} = \dots$

Answer (Please select your correct option)

☐ $-\frac{f_x}{f_y}$

☐ $-\frac{f_y}{f_x}$

☐ $\frac{f_x}{f_y}$

☐ $\frac{f_y}{f_x}$

Start Time: 6:56 AM

58:00

Time Left

5 

For $y^2(1-x^2) - \cos^2 x = c$ with $y(0) = 3$, the value of c is-----.

Answer (Please select your correct option)

☐ 9

☐ 8

correct answer solve
by hadi

☐ 10

☐ 5

The differential equation $(3x^2y + 2) dx + (x^3 + y) dy = 0$ is -----.

Answer (Please select your correct option)

☐ Exact

correct answer solve
by hadi

☐ Non-exact

☐ Separable

☐ Homogenous

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t$; the continuous function, $q(t)$, is.....

Answer (Please select your correct option)

☐ $2t$ ☐ $-\frac{1}{t}$ ☐ $\frac{1}{t}$ ☐ t

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min) 

For a 1st order linear differential equation $\frac{dy}{dx}(x + 2y^2) = y$; The continuous function, $q(y)$, is---

Answer (Please select your correct option)

- ☐ x^2
- ☐ $2y$
- ☐ $2y^2$
- ☐ $-2y$
- correct answer solve
by hadi

Start Time: 6:56 AM

57:00

Time Left

9 

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n$ for $n \neq 0, 1$ is called a/an ----.

Answer (Please select your correct option)

☐

Non-Exact differential equation

☐

Linear differential equation

☐

Bernoulli equation

correct answer solve
by hadi

☐

Exact differential equation

Made By: Waqar Siddhu

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are-----.

Answer (Please select your correct option)

☐

$P=0, P= b/a$

☐

$P=0, P=a/b$

correct answer solve
by hadi

☐

$P=a/b, P=b/a$

☐

$P=0, P=0$

The logistic equation $\frac{dP}{dt} = P(a - bP)$ is a/an-----.

Answer (Please select your correct option)

- ☐ Linear differential equation
- ☐ Non-linear differential equation
- ☐ Homogenous differential equation
- ☐ Exact differential equation

correct answer solve
by hadi

Wronskian of $\sin x, \cos x$ is----

Answer (Please select your correct option)

☐ $\sin x$

☐ 1

☐ -1

☐ $\cos x$

correct answer solve
by hadi

Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min) 

What is annihilator operator of the function $\frac{dy}{dx} + 4y = e^{2x}$?

Answer (Please select your correct option)

☐ $(D+4)(D-2)$ ☐ $(D-2)$ ☐ $(D+2)$ ☐ $D(D^2-2)$

Start Time: 6:56 AM

57:00

Time Left

14 



What is the annihilator operator of the function $g(x) = \cos 2x$?

Answer (Please select your correct option)

☐ $(D - 4)$

☐ $(D + 4)$

☐ $(D^2 + 4)$

☐ $(D^3 - 4)$

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

☐ $\frac{dy}{dx} = -\frac{x}{y}$

☐ $\frac{dy}{dx} = \frac{x}{y}$

☐ $\frac{dy}{dx} = -\frac{y}{x}$

☐ $\frac{dy}{dx} = \frac{y}{x}$

correct answer solve
by hadi

Made By: Waqar Siddhu



The orthogonal trajectory to the family of curves $3x + 4y = c$ is-----.

Answer (Please select your correct option)

☐

$$y = 2x$$

☐

$$y - x = c$$

☐

$$4x - 3y = c$$

correct answer solve
by hadi

☐

$$4x + 3y = c$$

Made By: Waqar Siddhu

In the population growth model, the solution of the differential equation is given by

Answer (Please select your correct option)

☐ $A(t) = ke^{kt}$

☐ $A(t) = A_0 e^{kt}$

correct answer solve
by hadi

☐ $A(t) = A_0 / 2$

☐ $A(t) = ke^{A_0 t}$

Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

If $y_2 = x^3$ is the second solution of $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- ☐ The first solution say y_1 must be non-trivial on the indicated interval.
- ☐ The first solution say y_1 can be trivial on the indicated interval.
- ☐ The first solution say y_1 must be discontinuous on the indicated interval.
- ☐ The first solution say y_1 may or may not be continuous on the indicated interval.

correct answer solve
by hadi

Start Time: 6:56 AM

57:00

Time Left



19



If the complementary solution of $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3\sin x$ is $c_1e^{-x} + c_2e^{2x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

☐ The general form of the particular solution will be $(A\sin x + B\cos x)e^x$.

☐ The general form of the particular solution will be $Ax\sin x + Bx\cos x$.

correct answer solve
by hadi

☐ The general form of the particular solution will be $(Ax\sin x + Bx\cos x)e^{-x}$.

☐ The general form of the particular solution will be $A\sin x + B\cos x$.

The solutions $y_1 = c_1$, $y_2 = c_2 \cos x$, $y_3 = c_3 \sin x$ of a Differential Equation are

Answer (Please select your correct option)

☐ Linearly independent

correct answer solve
by hadi

☐ Linearly dependent

☐ both of the other options

Wronskian of the functions $y_1(x) = 1$, $y_2(x) = \cos x$, $y_3(x) = \sin x$ is

Answer (Please select your correct option)

0

☒



1

☐

2

☐

3

☐



The value of amplitude in the solution $X=7\sin(3t+2.45)$ is

Answer (Please select your correct option)

☐

7

correct answer solve
by hadi

☐

3

☐

2.45

☐

1

Made By: Waqar Siddhu



The order of the differential equation $\left(\frac{du}{dt}\right)^{\frac{1}{3}} = \frac{1}{u}$ is

Answer (Please select your correct option)



1

correct answer solve
by hadi



2



3



4

Made By: Waqar Siddhu

The solution of the differential equation $\frac{dy}{dx} = \tan x$ is

Answer (Please select your correct option)

☐ $y = \ln |\sin x| + c$

☐ $y = \ln |\sec x| + c$

☐ $y = \ln |\operatorname{cosec} x| + c$

correct answer solve
by hadi

☐ $y = \ln |\cos x| + c$

A differential equation $M(x, y) dx + N(x, y) dy = 0$ is said to be an exact if -----.

Answer (Please select your correct option)

☐ $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

☐ $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

☐ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

☐ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

correct answer solve
by hadi

Made By: Waqar Siddhu



The differential equation $(x^2 - 2x + 2y^2) dx + 2xy dy = 0$ is a/an -----differential equation.

Answer (Please select your correct option)

Exact

☐

Non-exact

☐

Homogenous

☐

Separable

☐

correct answer solve
by hadi

Made By: Waqar Siddhu

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t^2$; the integrating factor will be -----.

Answer (Please select your correct option)

☐ $\mu = -\frac{1}{t}$

☐ $\mu = \frac{1}{t}$

correct answer solve
by hadi

☐ $\mu = t$

☐ $\mu = 2t$

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n$ for $n \neq 0, 1$ is called a/an ----.

Answer (Please select your correct option)

☐ Non-Exact differential equation

☐ Linear differential equation

☐ Bernoulli equation

☐ Exact differential equation

correct answer solve
by hadi

Made By: Waqar Siddhu

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^x$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)

☐ $v = y^{x-1}$

☐ $v = y^{1-x}$

☐ $v = y^x$

☐ $v = y'$

correct answer solve
by hadi

To convert the non-homogeneous differential equation $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$ into separable form, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then we use the substitution-----.

Answer (Please select your correct option)

☐ $x = X + h$

☐ $y = Y + h$

☐ $z = a_1x + b_1y$

☐ $z = Y + X$

correct answer solve
by hadi

Made By: Waqar Siddhu

In exponential model for the population growth $P(t) = P_0 e^{kt}$ If $k > 0$, then $\lim_{t \rightarrow \infty} P(t) = \dots$

Answer (Please select your correct option)

☐ ∞

correct answer solve
by hadi

☐ $-\infty$ ☐

0

☐

1

Made By: Waqar Siddhu



The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are-----.

Answer (Please select your correct option)

☐

$P=0, P= b/a$

☐

$P=0, P=a/b$

correct answer solve
by hadi

☐

$P=a/b, P=b/a$

☐

$P=0, P=0$

The orthogonal trajectory to the family of curves $3x + 4y = c$ is-----.

Answer (Please select your correct option)

☐ $y = 2x$

☐ $y - x = c$

☐ $4x - 3y = c$

☐ $4x + 3y = c$

correct answer solve
by hadi

Made By: Waqar Siddhu



If the tangent lines of two curves are perpendicular at their point of intersection then both the curves are-----

Answer (Please select your correct option)



Non-intersecting curves



Parallel curves



Orthogonal curves

correct answer solve
by hadi



Intersecting curves

Made By: Waqar Siddhu



If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that $K=0.043$?

Answer (Please select your correct option)



371.415



380.560



363.279

correct answer solve
by hadi



360.351

Made By: Waqar Siddhu



If $y = 1 + 6x^2 - 7x^3$, then which of the following is true for it ?

Answer (Please select your correct option)

☐

Its annihilator operator is D .

☐

Its annihilator operator is D^2 .

☐

Its annihilator operator is D^3 .

☐

Its annihilator operator is D^4 .

correct answer solve
by hadi

Made By: Waqar Siddhu

If $y = e^{-6x}$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^2$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^3$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D + 6)$.

correct answer solve
by hadi

Made By: Waqar Siddhu

If $y_1 = x$ and $y_2 = xe^x$ are the first and second solution of $x^2 \frac{d^2 y}{dx^2} - x(x+2) \frac{dy}{dx} + (x+2)y = 0$ on $(0, \infty)$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly dependent.

☐ $y_1 = x$ and $y_2 = xe^x$ must be linearly independent.

correct answer solve
by hadi

☐ $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly independent.

☐ $y_1 = x$ and $y_2 = xe^x$ must be linearly dependent.

If $y_2 = x^3$ is the second solution of $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ The first solution say y_1 must be non-trivial on the indicated interval.

☐ The first solution say y_1 can be trivial on the indicated interval.

correct answer solve
by hadi

☐ The first solution say y_1 must be discontinuous on the indicated interval.

☐ The first solution say y_1 may or may not be continuous on the indicated interval.



A solution obtained by giving particular values to the arbitrary constants in the General Solution of a differential equation is called a -----

Answer (Please select your correct option)

☐ Singular solution

☐ Particular solution

correct answer solve
by hadi

☐ Explicit Solution

☐ None of these

Made By: Waqar Siddhu

The equation $\frac{dy}{dx} = \frac{x(x+1)}{y(y-1)}$ is a/an -----

Answer (Please select your correct option)

☐ partial differential equation.

i think this ...because
in partial all value
write up and down ,,,

☐ ordinary differential equation.

☐ polynomial equation.

☐ transcendental equation.

Made By: Waqar Siddhu



A differential equation $M(x, y) dx + N(x, y) dy = 0$ is said to be non-exact if -----.

Answer (Please select your correct option)

☐ $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

☐ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

☐ $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

☐ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t^2$; the integrating factor will be -----.

Answer (Please select your correct option)

☐ $\mu = -\frac{1}{t}$

☐ $\mu = \frac{1}{t}$

correct answer solve
by hadi

☐ $\mu = t$

☐ $\mu = 2t$

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)



$$v = y^{n-1}$$



$$v = y^{1-n}$$

correct answer solve
by hadi



$$v = y^n$$



$$v = y'$$

Made By: Waqar Siddhu

The differential equation $\frac{dy}{dx} = \frac{x+2y-4}{2x+y-5}$ is not homogeneous. To convert it into homogeneous form, we use the substitution-----.

Answer (Please select your correct option)

☐

$$x = hX$$

☐

$$y = kY$$

☐

$$z = x+2y$$

correct answer solve
by hadi...but check it

☐

$$x = X+h, y = Y+k$$

Made By: Waqar Siddhu

Number of constant solutions obtained from the separable differential equation $\frac{dy}{dx} = \frac{y^3 - 1}{x}$ is---

Answer (Please select your correct option)

☐ 0 correct answer solve by hadi

☐ 1

☐ 2

☐ 3

Made By: Waqar Siddhu



In the 1st order linear differential equation, $\frac{dy}{dx} + \frac{y}{\ln x} = x^2$ find $p(x)$.

Answer (Please select your correct option)

☐ $\frac{y}{\ln x}$

☐ $\frac{1}{\ln x}$

☐ x^2

☐ $\frac{x^2}{\ln x}$



For a 1st order linear differential equation $\frac{dy}{dx} + y = x$, the integrating factor $\mu(x)$ is---

Answer (Please select your correct option)

☐ x

☐ 1

correct answer solve
by hadi

☐ e^x

☐ e

Made By: Waqar Siddhu



In exponential model for the population growth $P(t) = P_0 e^{kt}$ If $k < 0$, then $\lim_{t \rightarrow \infty} P(t) = \dots$

Answer (Please select your correct option)

☐ ∞

☐ 0

☐ $-\infty$

☐ 1

correct answer solve
by hadi

Made By: Waqar Siddhu



What is other name of Logistic Model?

Answer (Please select your correct option)

☐ Epidemic Model

☐ Euler Model

☐ Verhulst Pearl Model

☐ Gussain Model

correct answer solve
by hadi

Made By: Waqar Siddhu



Wronskian of $\sin x, \cos x$ is----

Answer (Please select your correct option)

☐ $\sin x$

☐ 1

☐ -1

☐ $\cos x$

correct answer solve
by hadi

Made By: Waqar Siddhu



What is annihilator operator of the function $\frac{dy}{dx} + 4y = e^{2x}$?

Answer (Please select your correct option)

☐ $(D+4)(D-2)$

☐ $(D-2)$

☐ $(D+2)$

☐ $D(D^2-2)$

correct answer solve
by hadi

Made By: Waqar Siddhu



What is the annihilator operator of the function $g(x) = \cos 2x$?

Answer (Please select your correct option)



$(D - 4)$



$(D + 4)$



$(D^2 + 4)$

i think this



$(D^3 - 4)$

Made By: Waqar Siddhu

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

☐ $\frac{dy}{dx} = -\frac{x}{y}$

☐ $\frac{dy}{dx} = \frac{x}{y}$

☐ $\frac{dy}{dx} = -\frac{y}{x}$

☐ $\frac{dy}{dx} = \frac{y}{x}$

correct answer solve
by hadi

Made By: Waqar Siddhu

If T denotes the half life and the initial amount of the substance be A_0 , then the amount of radioactive substance at time T is given by-----.

Answer (Please select your correct option)



$$A(T) = \frac{A_0}{2}$$

correct answer solve
by hadi



$$A(T) = \frac{T}{2}$$



$$A(T) = A_0$$



$$A(T) = \frac{kT}{2}$$

Made By: Waqar Siddhu



If the auxiliary equation has complex roots $\pm i$, then the general solution of the differential equation is _____

Answer (Please select your correct option)

☐ $y = c_1 \cos x + c_2 \sin x$

☐ $y = ce^{2x} (\cos x + \sin x)$

☐ $y = e^{ix} (c_1 \cos x + c_2 \sin x)$

correct answer solve
by hadi...but check it

☐ $y = e^x (c_1 \cos x - c_2 \sin x)$

Made By: Waqar Siddhu

If the complementary solution of $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 4e^{2x}$ is $c_1e^{2x} + c_2e^{3x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

☐ The general form of the particular solution will be Ae^{3x} .

☐ The general form of the particular solution will be Ae^x .

☐ The general form of the particular solution will be Ae^{2x} .

☐ The general form of the particular solution will be Axe^{2x} .



If $c_1 e^x$ and $c_2 x e^x$ are the solutions of $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0$, then which of the following is the most accurate option for $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = e^x$?

Answer (Please select your correct option)

☐ Its general form of the particular solution will be $A e^x$.

☐ Its general form of the particular solution will be $A x e^x$.

☐ Its general form of the particular solution will be $A x^2 e^x$.

☐ Its general form of the particular solution will be $A x^3 e^x$.



If $e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$ is the solution of $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 13y = 0$, then which of the following is the most accurate option for $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 13y = e^{2x} \sin 3x$?

Answer (Please select your correct option)



Its general form of the particular solution will be $Ae^{2x} + B \sin x + C \cos x$.



Its general form of the particular solution will be $e^{2x}(A \sin x + B \cos x)$.



Its general form of the particular solution will be $e^{2x}(Ax \sin x + Bx \cos x)$.

correct answer solve
by hadi.h



Its general form of the particular solution will be $e^{2x}(Ax \sin 3x + Bx \cos 3x)$.

Made By: Waqar Siddhu

Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min) 

The form of the exact solution to

$$2\frac{dy}{dx} + 3y = e^{-x}, y(0) = 5$$

is

Answer (Please select your correct option)

☐ $Ae^{-1.5x} + Bxe^{-x}$

☐ $Ae^{1.5x} + Be^{-x}$

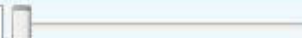
☐ $Ae^{1.5x} + Bxe^{-x}$

☒ $Ae^{-1.5x} + Be^{-x}$

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1 

Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min) 

The Period of oscillator in the solution $X=50\sin(20t+8.5)$ is

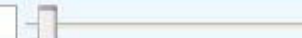
Answer (Please select your correct option)

☐ 0.17643☐ 0.32045☐ 0.31400☐ 0.58000

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2 

Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

The equation $\frac{dy}{dx} = \frac{x(x+1)}{y(y-1)}$ is a/an

Answer (Please select your correct option)

☐ partial differential equation.☒ ordinary differential equation.

i guess

☐ polynomial equation.☐ transcendental equation.

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Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

If $f(x,y)=0$, then $\frac{dy}{dx} = \dots$

Answer (Please select your correct option)

☐ $-\frac{f_x}{f_y}$

☐ $-\frac{f_y}{f_x}$

☐ $\frac{f_x}{f_y}$

☐ $\frac{f_y}{f_x}$

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Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

For $y^2(1-x^2) - \cos^2 x = c$ with $y(0) = 3$, the value of c is -----.

Answer (Please select your correct option)

☐ 9☒ 8☐ 10☐ 5

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Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

The differential equation $(3x^2y + 2) dx + (x^3 + y) dy = 0$ is -----.

Answer (Please select your correct option)

☒ Exact☐ Non-exact☐ Separable☐ Homogenous

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Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min) 

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t$; the continuous function, $q(t)$, is.....

Answer (Please select your correct option)

☒ $2t$ ☐ $-\frac{1}{t}$ ☐ $\frac{1}{t}$ ☐ t

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Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min) 

For a 1st order linear differential equation $\frac{dy}{dx}(x + 2y^2) = y$; The continuous function, $q(y)$, is---

Answer (Please select your correct option)

 x^2  $2y$  $2y^2$  $-2y$

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9 

Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min) 

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n$ for $n \neq 0, 1$ is called a/an ----.

Answer (Please select your correct option)

☐ Non-Exact differential equation☐ Linear differential equation☒ Bernoulli equation☐ Exact differential equation

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Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min) 

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are-----

Answer (Please select your correct option)

☐ P=0, P= b/a☒ P=0, P=a/b☐ P=a/b, P=b/a☐ P=0, P=0

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Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

Wronskian of $\sin x, \cos x$ is-----.

Answer (Please select your correct option)

- ☐ $\sin x$
- ☐ 1
- ☒ -1
- ☐ $\cos x$

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Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min) 

What is annihilator operator of the function $\frac{dy}{dx} + 4y = e^{2x}$?

Answer (Please select your correct option)

☐ $(D+4)(D-2)$ ☐ $(D-2)$ ☐ $(D+2)$ ☐ $D(D^2-2)$

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Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min) 

What is the annihilator operator of the function $g(x) = \cos 2x$?

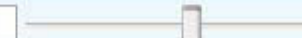
Answer (Please select your correct option)

- ☐ $(D-4)$
- ☐ $(D+4)$
- ☐ (D^2+4)
- ☐ (D^3-4)

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Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

☐ $\frac{dy}{dx} = -\frac{x}{y}$

☐ $\frac{dy}{dx} = \frac{x}{y}$

☐ $\frac{dy}{dx} = -\frac{y}{x}$

☒ $\frac{dy}{dx} = \frac{y}{x}$

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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min) 

The orthogonal trajectory to the family of curves $3x + 4y = c$ is-----.

Answer (Please select your correct option)

☐ $y = 2x$

☐ $y - x = c$

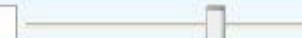
☒ $4x - 3y = c$

☐ $4x + 3y = c$

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17 

Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

In the population growth model, the solution of the differential equation is given by

Answer (Please select your correct option)

☐ $A(t) = ke^{kt}$

☒ $A(t) = A_0 e^{kt}$

☐ $A(t) = A_0 / 2$

☐ $A(t) = ke^{A_0 t}$

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Question No : 19 of 26

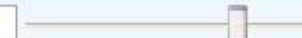
Marks: 1 (Budgeted Time 1 Min) 

If $y_2 = x^3$ is the second solution of $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- ☐ The first solution say y_1 must be non-trivial on the indicated interval.
- ☐ The first solution say y_1 can be trivial on the indicated interval.
- ☐ The first solution say y_1 must be discontinuous on the indicated interval.
- ☐ The first solution say y_1 may or may not be continuous on the indicated interval.

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Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

If the complementary solution of $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3\sin x$ is $c_1 e^{-x} + c_2 e^{2x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

☐ The general form of the particular solution will be $(A \sin x + B \cos x) e^x$.

☒ The general form of the particular solution will be $Ax \sin x + Bx \cos x$.

☐ The general form of the particular solution will be $(Ax \sin x + Bx \cos x) e^{-x}$.

☐ The general form of the particular solution will be $A \sin x + B \cos x$.

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20



In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

into linear differential equation, we choose ----.

Choices:

☐ $v = y^{n-1}$

☒ $v = y^{1-n}$

☐ $v = y^n$

☐ $v = y'$

TIME LEFT

8



Differential Equations (MTH401)

Question: 2 (Marks: 1)

Attempted Questions: 25 Total Questions: 26

If $y = x$, then $\frac{dy}{dx} = \dots\dots\dots$

Choices:

0

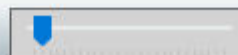


1

 x  $x+1$ 

TIME LEFT

8



Next Question



The orthogonal trajectory to the family of curves $3x + 4y = c$ is-----.

Choices:

☐ $y = 2x$

☐ $y - x = c$

☒ $4x - 3y = c$

☐ $4x + 3y = c$

TIME LEFT

8

If the auxiliary equation of a differential equation has two distinct real roots m_1 and m_2 , then the general solution of the differential equation is _____

Choices:

☒ $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

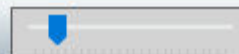
☐ $y = c_1 e^{m_1 x} + c_2 e^{-m_2 x}$

☐ $y = c_1 e^{-m_1 x} + c_2 e^{m_2 x}$

☐ $y = c_1 e^{m_1 + m_2 x}$

TIME LEFT

8



If the annihilator operator of $2x$ is D^2 and of xe^{3x} is $(D-3)^2$, then which of the following is the correct option?

Choices:

☐ The annihilator operator of $2x + xe^{3x}$ is $(D-3)^2$.

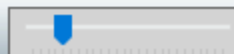
☐ The annihilator operator of $2x + xe^{3x}$ is D^2 .

☐ The annihilator operator of $2x + xe^{3x}$ is $D^2 + (D-3)^2$.

☒ The annihilator operator of $2x + xe^{3x}$ is $D^2 (D-3)^2$.

TIME LEFT

8



$$\frac{dy}{dx} = \frac{1}{x}$$

If _____, then which of the following is the most appropriate option for it?

Choices:

☒ $y = \ln x + c$

☐ $y = x^2$

☐ $y = \ln \sqrt{x} + c$

☐ $y = \ln \frac{1}{\sqrt{x}} + c$

TIME LEFT

8



$$\frac{dy}{dx} = \frac{2x + 3y - 4}{4x + 6y + 9}$$

If _____, then which of the following substitution is appropriate to make it a homogeneous differential equation?

Choices:

☐ $z = 2x + 3y$

☐ $z = x + y$

☐ $z = 4x + 3y$

☐ $z = 2x + 6y$

TIME LEFT

8

For a homogeneous differential equation of the form $y f(xy)dx + xg(xy)dy = 0$ and if $xM - yN \neq 0$ then the Integrating factor is.....

Choices:

☐ $\mu = \frac{1}{xM + yN}$

☒ $\mu = \frac{1}{xM - yN}$

☐ $\mu = \frac{1}{yM + xN}$

☐ None of these

TIME LEFT

8



How many initial conditions are necessary to determine the particular solution of a **first order linear** differential equation?

Choices:

0



1



2



3



TIME LEFT

8



If $y_1 = xe^{-x}$ is the first solution of $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$, then which of the following is true for

$$y_2 = y_1 \int \frac{e^{-2x}}{y_1^2}$$

its second solution

Choices:

☐ $xe^{-x} \int \frac{2}{e^{-2x}} dx$

☐ $xe^{-x} \int \frac{2}{x^2 e^{-2x}} dx$

☐ $xe^{-x} \int \frac{1}{x^2 e^{-2x}} dx$

☒ $xe^{-x} \int \frac{1}{x^2} dx$

TIME LEFT

8

Differential Equations (MTH401)

Question: 11 (Marks: 1)

Attempted Questions: 25 Total Questions: 26

Wronskian $W(x, x^2)$ is----

Choices:

☐ x☒ x^2 ☐ x^3 ☐ 0

TIME LEFT

8



$$\frac{dy}{dx} + p(x)y = q(x)$$

In first order linear differential equation , the functions $p(x)$ and $q(x)$ must be

Choices:

Continuous



Discontinuous



Homogeneous



Non homogeneous



TIME LEFT

8

Differential of $(x+y)$ is -----.

Choices:

☒ $dx + dy$

☐ $dx - dy$

☐ $d(xy)$

☐ $d\left(\frac{x}{y}\right)$

TIME LEFT

7



Which of following is an example of boundary value problem (BVP) ?

Choices:

☐ $\frac{d^2y}{dx^2} + \lambda^2 x = 0, \quad x(0) = x_0, x'(0) = 0$

☒ $\frac{d^2y}{dx^2} + \lambda^2 x = 0, \quad x(0) = x_0, x'(1) = 0$

☐ $\frac{d^2y}{dx^2} + \lambda^2 x = 0, \quad x(0) = x_0$

☐ $\frac{d^2y}{dx^2} + \lambda^2 x = 0, \quad x'(1) = 0$

TIME LEFT

7

For a 2nd order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 3, W_1 = -\frac{1}{4}$, then $u_1' = \dots$.

Choices:

12



-12

 $\frac{1}{12}$  $-\frac{1}{12}$ 

TIME LEFT

7



Next Question



If $y = a \sin x + b \cos x$ is the complementary solution of $y'' + y = 1$, then the general form of its particular solution is $y_p = \dots$

Choices:

☒ A ☐ Ax ☐ $Ax + B$ ☐ $Ax \sin x + Bx \cos x$

TIME LEFT

7

For some given functions, the Wronskian $W \neq 0$ somewhere in given interval, then given functions are linearly -----.

Choices:

Independent



dependent



TIME LEFT

7

The value of amplitude in the solution of simple harmonic motion

$$x(t) = 40 \sin(7t + 1.86)$$

is.....

Choices:

7



15



40



20



TIME LEFT

7



The orthogonal trajectory to the family of curves $x + 2y = 2$ is-----.

Choices:

☐ $y = -2x$

☐ $y - 2x = c$

☐ $2x - 3y = c$

☒ $4x + 3y = c$

TIME LEFT

7

If the tangent lines of two curves are perpendicular at their point of intersection then both are-----

Choices:

Orthogonal



Non-Orthogonal



TIME LEFT

7

